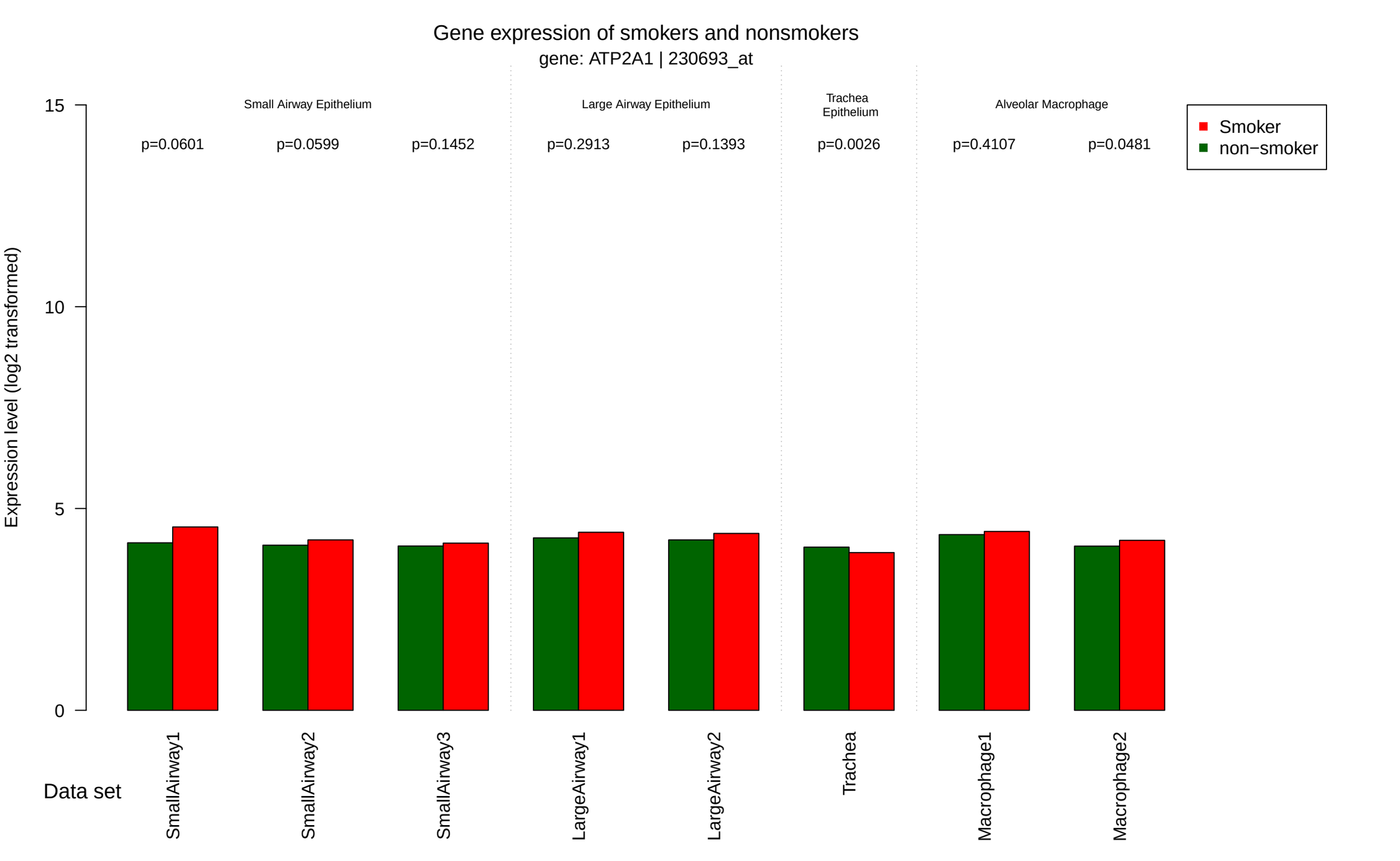
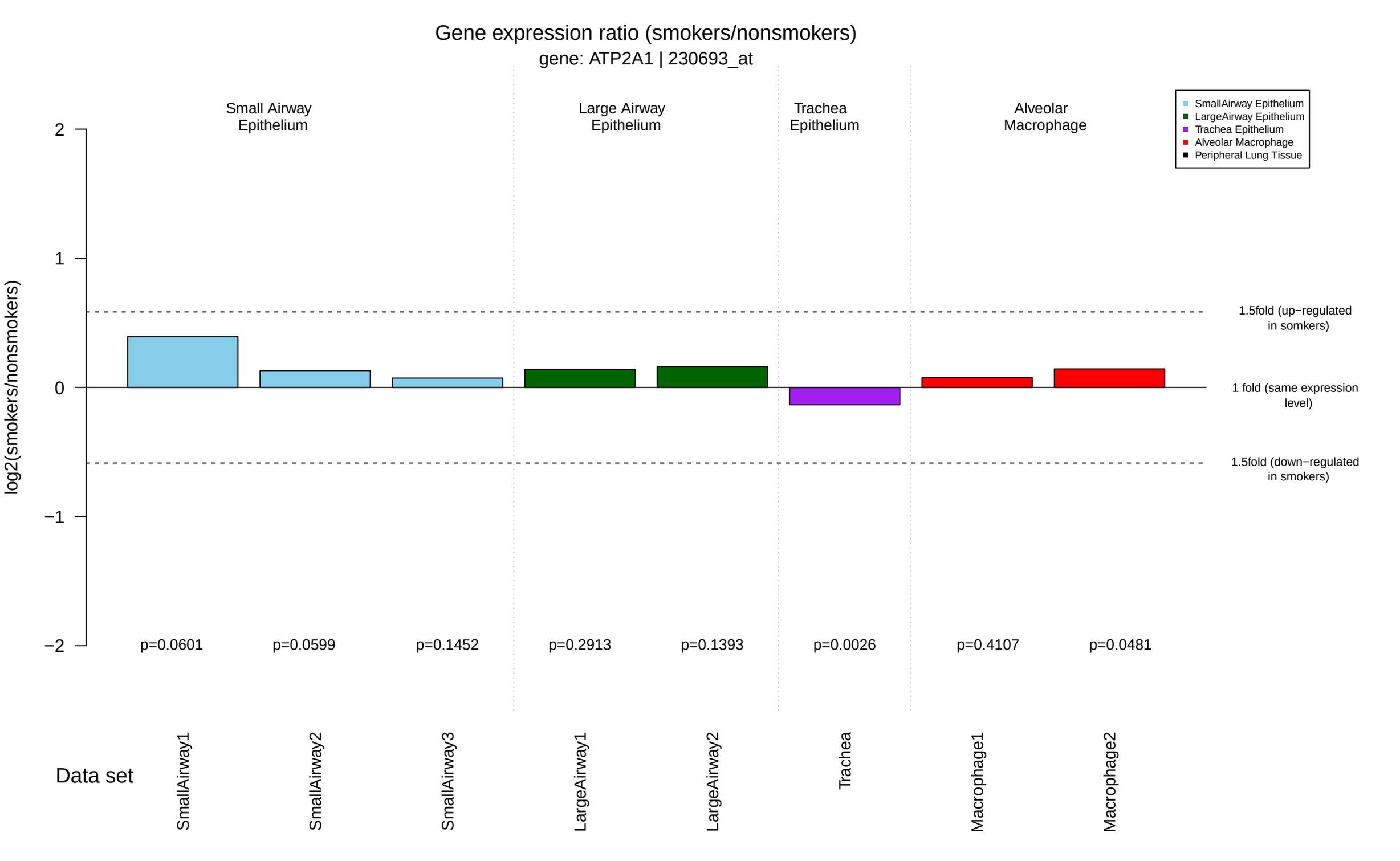
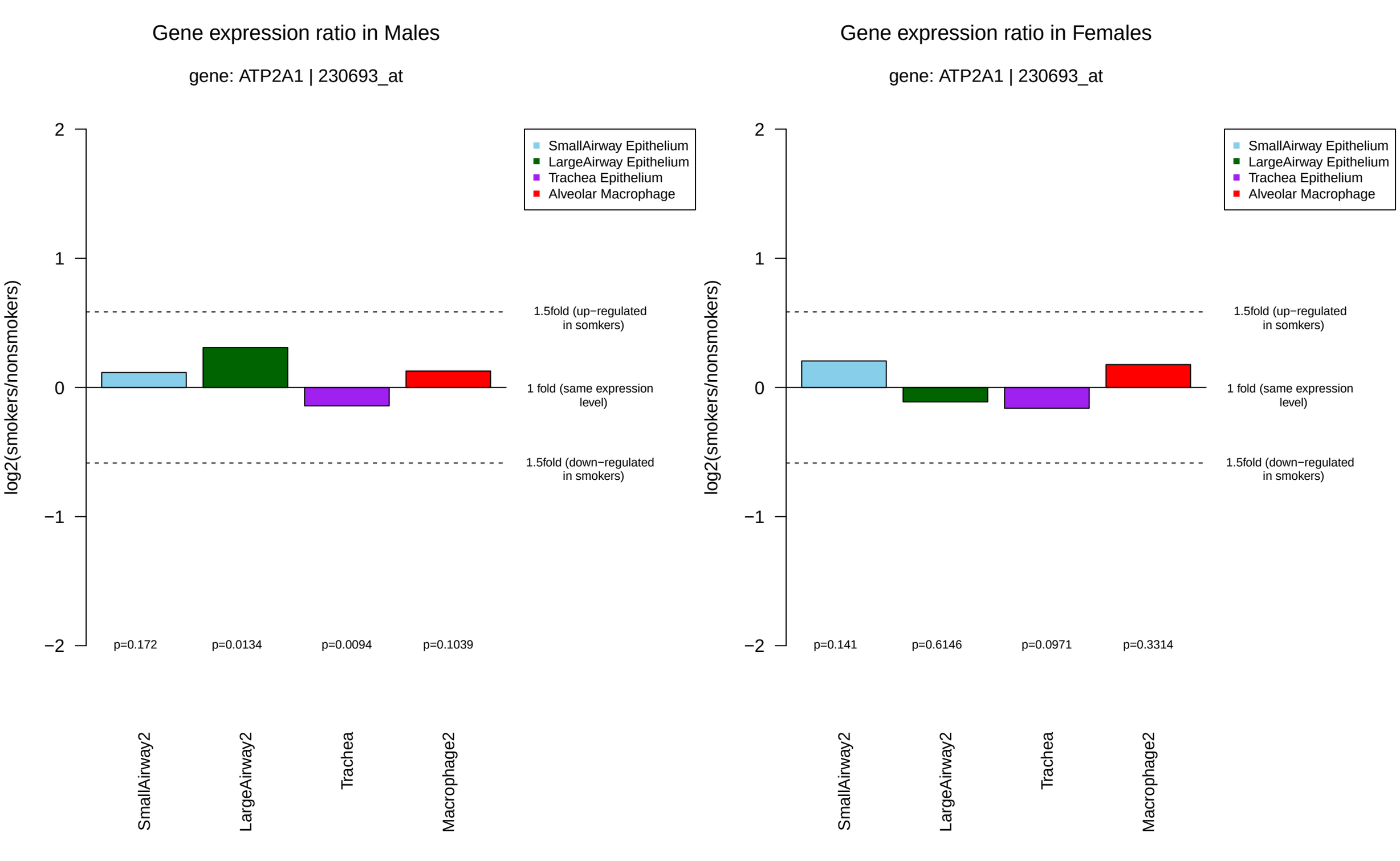
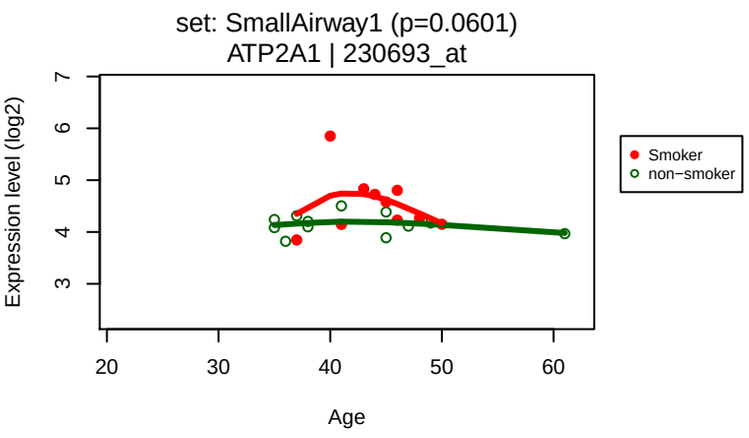
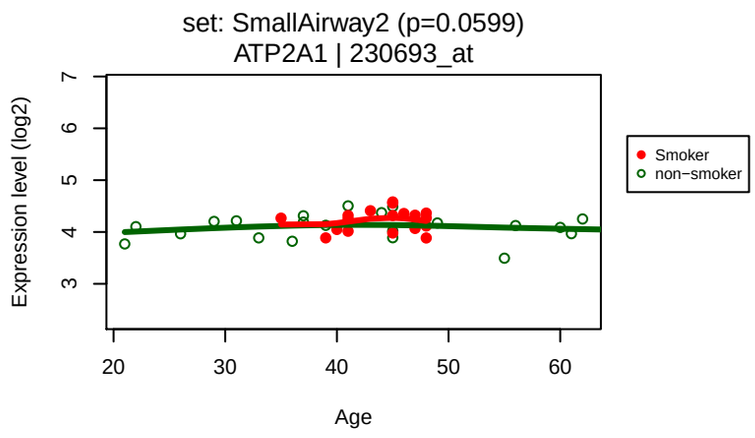
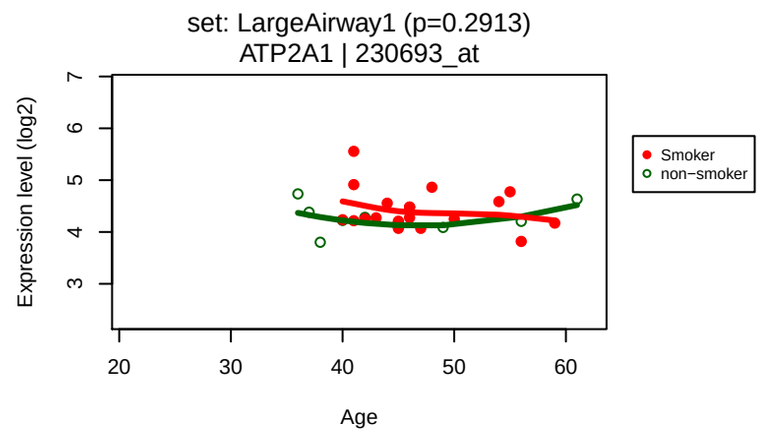

Gene expression of smokers and nonsmokers
gene: ATP2A1 | 230693_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0601
p=0.0599
p=0.1452
p=0.2913
p=0.1393
p=0.0026
p=0.4107
p=0.0481
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

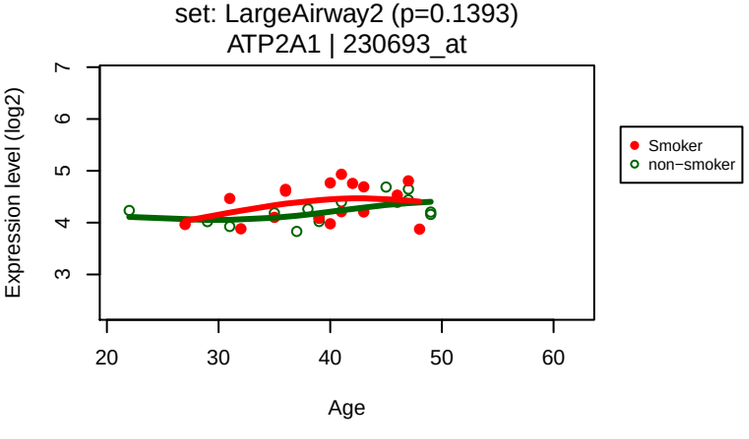

Gene expression ratio (smokers/nonsmokers)
gene: ATP2A1 | 230693_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0601
p=0.0599
p=0.1452
p=0.2913
p=0.1393
p=0.0026
p=0.4107
p=0.0481
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

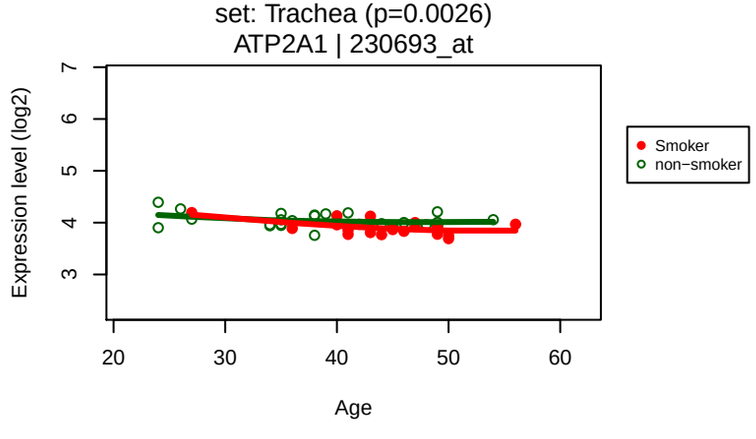

Gene expression ratio in Males
gene: ATP2A1 | 230693_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.172
p=0.0134
p=0.0094
p=0.1039
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2A1 | 230693_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.141
p=0.6146
p=0.0971
p=0.3314
SmallAirway2
LargeAirway2
Trachea
Macrophage2

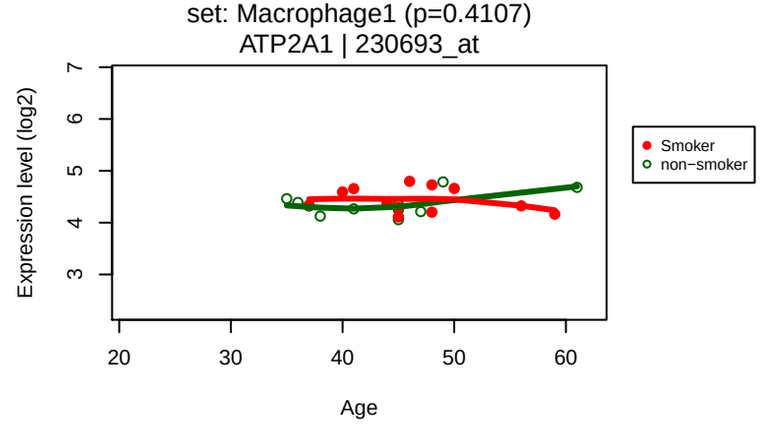

set: SmallAirway1 (p=0.0601)
ATP2A1 | 230693_at
Expression level (log2)
Age
Smoker
non−smoker

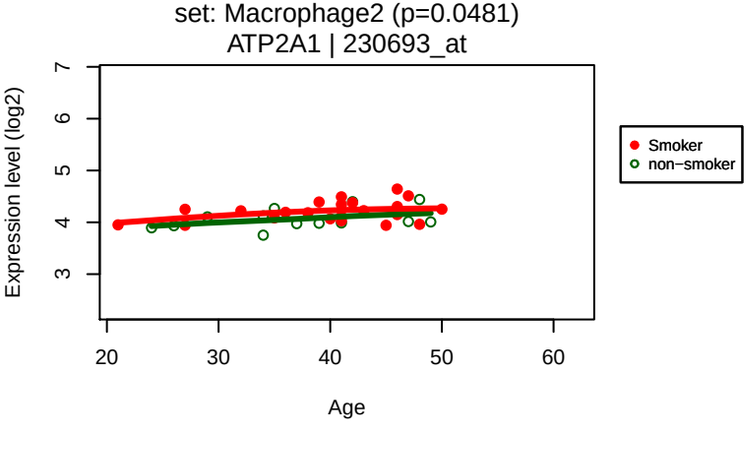

set: SmallAirway2 (p=0.0599)
ATP2A1 | 230693_at
Expression level (log2)
Age
Smoker
non−smoker

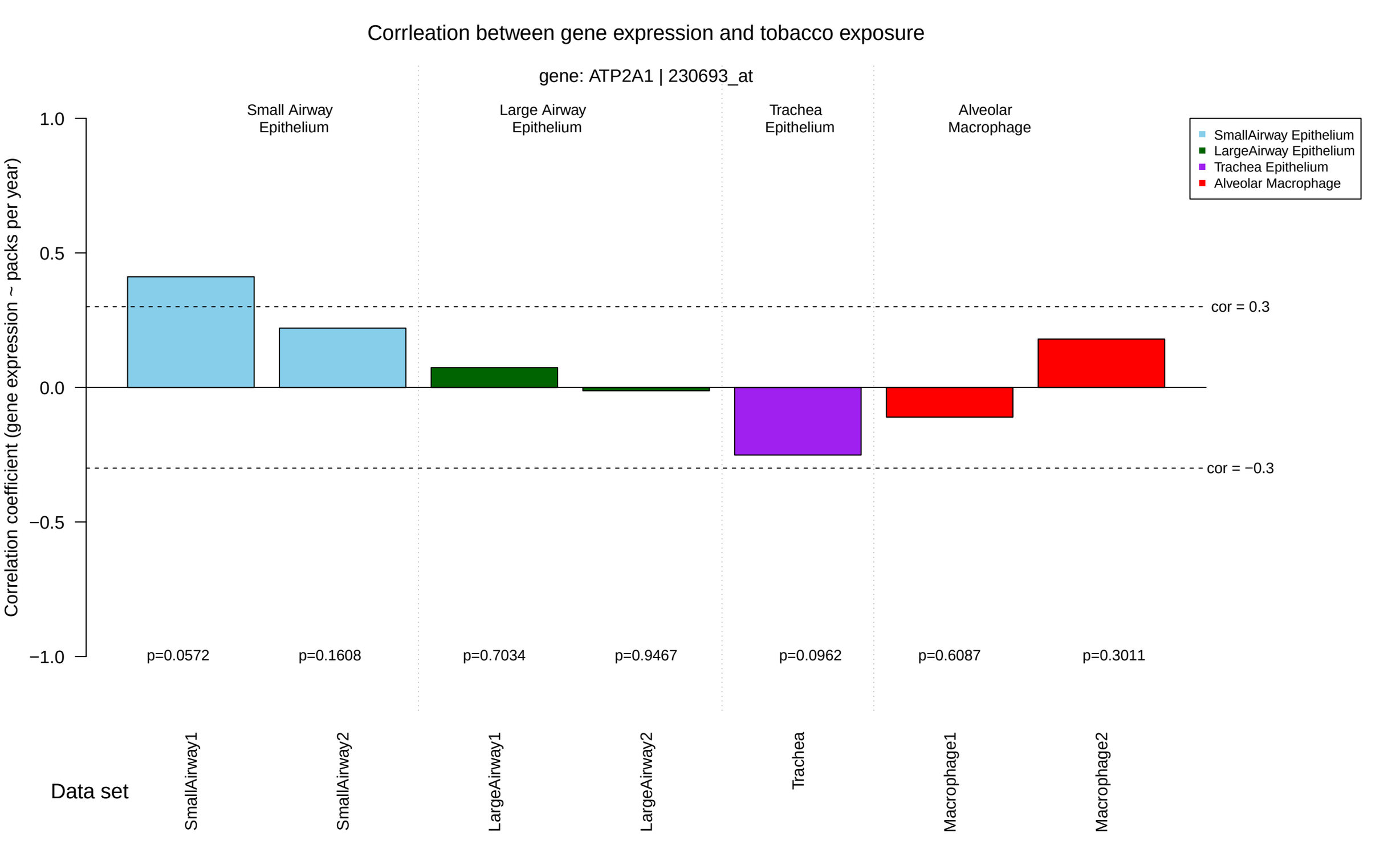

set: LargeAirway1 (p=0.2913)
ATP2A1 | 230693_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1393)
ATP2A1 | 230693_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0026)
ATP2A1 | 230693_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4107)
ATP2A1 | 230693_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0481)
ATP2A1 | 230693_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2A1 | 230693_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0572
p=0.1608
p=0.7034
p=0.9467
p=0.0962
p=0.6087
p=0.3011
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2